

From dozens of SQL tables to digital biomarker features

Every raw table must be transformed before it can become a clinically interpretable feature family.

SQL tables tracked

44

Reviewed tables

10

Candidate features

69

Selected features

29

Each SensorDB table needs a dedicated extraction algorithm
before it can contribute reliable model features.

This means table-by-table SQL logic, row interpretation, deduplication, privacy handling, feature calculation, and validation.

All SQL tables

• accelerometer	• network_traffic
✓ applications_foreground	• plugin_google_activity_recognition
• aware_device	• proximity
• aware_log	• rotation
• aware_studies	• screen
• barometer	• sensor_accelerometer
✓ battery	• sensor_barometer
• battery_charges	• sensor_bluetooth
• battery_discharges	• sensor_gravity
✓ bluetooth	• sensor_gyroscope
✓ calls	• sensor_light
• gravity	• sensor_linear_accelerometer
✓ gsm	• sensor_magnetometer
• gsm_neighbor	• sensor_proximity
• gyroscope	• sensor_rotation
✓ keyboard	• sensor_temperature
✓ light	• sensor_wifi
• linear_accelerometer	• significant
✓ locations	• telephony
• magnetometer	• timezone

Selected features already

Grouped by SQL table; these are aggregate features selected

applications_foreground • app_foreground_event_count • unique_foreground_apps • app_use_diversity
battery • low_battery_event_count • charging_or_plugged_event_count
bluetooth • unique_bluetooth_addresses • bluetooth_address_diversity_ratio
calls • call_event_count • incoming_call_count • outgoing_call_count • missed_rejected_blocked_call_count • total_call_duration_seconds
gsm • unique_gsm_cell_count • unique_gsm_lac_count • gsm_cell_transition_count
keyboard • keyboard_median_inter_event_interval_ms • keyboard_inter_event_interval_iqr_ms • keyboard_long_pause_count_2s • keyboard_typing_burst_count • keyboard_median_word_completion_time_ms • keyboard_deletion_event_count
light • median_light_lux • percent_dark_samples • night_mean_light_lux • light_lux_iqr

